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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech (VLSI Design & Embedded Systems) (CBCS) Degree Examinations

May - June 2024

22VDE201 – ANALOG VLSI DESIGN

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I

Marks BT* CO* PO*

1. a) For a NMOSFET operating in saturation, derive the expression for output resistance. Also draw the small signal model of a NMOSFET with $\gamma = 0$ and $\lambda \neq 0$. Determine the width of a NMOS current source with $I_D = 0.5$ mA operating with $V_{DS} = 0.4$ V. The minimum required r_o is 20 k Ω , $\lambda = 0.1$ V⁻¹ with $L = 0.5$ μ m and $V_{tn} = 0.7$ V.
- b) Analyze the phenomenon in connection with the slope of I_D versus V_{DS} characteristic of a NMOSFET operating in saturation. For the circuit shown in Figure Q.1b, plot V_{out} as a function of V_{in} as V_{in} varies from 0 to 3V. Assume $\gamma = 0$, $\lambda = 0$ and $V_{tn} = 0.7$ V.

10 L*3 1 2

10 L3 1 2

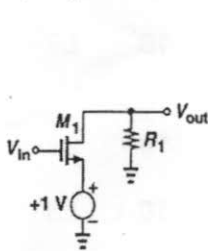


Figure Q.1b

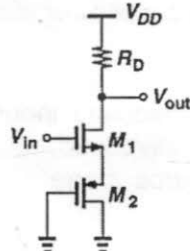


Figure Q.3a

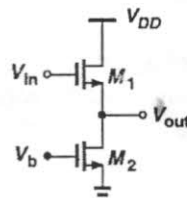


Figure Q.3b

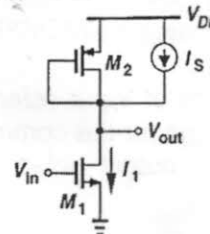


Figure Q.4a

2. a) For a NMOSFET operating in saturation, derive the expression for transconductance. A MOSFET carries a drain current of 1 mA with $V_{DS} = 0.5$ V in saturation.
 - i) Determine the change in I_D if V_{DS} rises to 1 V and $\lambda = 0.1$ V⁻¹.
 - ii) Determine the device output impedance, r_o .
- b) Write the complete small signal model of NMOSFET with $\gamma \neq 0$ and $\lambda \neq 0$. For a NMOSFET device, calculate the resistance of the device in linear region and the device output impedance in saturation region given that $\mu_n C_{ox} = 50$ μ A/V², $(W/L) = 10$, $V_{GS} = 2.7$ V and $V_{th} = 0.7$ V, $\lambda = 0.1$ V⁻¹ and $V_{DS} = 3$ V.

10 L3 1 2

10 L3 1 2

Unit – II

3. a) Derive the expression for the voltage gain of common source amplifier with source degeneration. Assume $\gamma = 0$ and $\lambda = 0$. Also determine the gain by inspection of the circuit shown in Figure Q.3a if $\lambda = 0$.
- b) Determine the voltage gain of a source follower. In the circuit shown in Figure Q.3b, find the minimum value of $(W/L)_2$ for which M_2 remain saturated. Given $I_2 = 200$ μ A, $(W/L)_1 = 20/0.5$, $V_{out} = 0.119$ V, $V_{tn} = 0.635$ V, $\lambda = 0.1$ V⁻¹, $\mu_n C_{ox} = 50$ μ A/V².
4. a) Draw the small signal model of a common source amplifier with diode connected PMOS load assuming $\lambda = 0$ and derive the expression for the voltage gain. In the circuit shown in Figure Q.4a (refer overleaf), M_1 is biased in saturation and $I_s = 0.75 I_1$, determine the small signal voltage gain, A_v .
- b) Determine the expression for voltage gain of the circuit shown in Figure Q.4b. Also determine (W/L) and g_m of M_1 for a drain current of 1 mA and a voltage gain of 0.8. Assume $\mu_n C_{ox} = 100$ μ A/V², gate overdrive voltage = 0.433 V, $R_s = 867$ Ω and $\lambda = 0$.

10 L3 2 3

10 L3 2 3

10 L3 2 3

10 L3 2 3

P.T.O

Unit – III

5. a) Derive the expression for the common mode voltage gain of a differential pair with resistive load driven by the common input voltage. Calculate the differential gain of a differential pair with $V_{DD} = 3\text{ V}$, $R_{D1} = R_{D2} = 3.16\text{ k}\Omega$, $I_{Tail} = 1\text{ mA}$ with source resistance $R_{ss} = 500\Omega$. Given: $\mu_n C_{ox} = 50\text{ }\mu\text{A/V}^2$, $(W/L)_{1,2} = 25/0.5$. 10 L3 3 3
- b) Explain the current copying in a current mirror circuit. Figure Q. 5b shows an arrangement where M1 and M2 serve as current sources for circuit 1 and circuit 2. Design the circuit for a power budget of 3 mW 10 L3 3 3
6. a) Starting from a variable gain amplifier, build a Gilbert cell. 10 L3 3 3
- b) Derive the expression for the differential gain of the circuit shown in Figure Q.6b using half-circuit concept. Assume $\lambda \neq 0$. 10 L3 3 3

Unit – IV

7. a) For a source follower, draw the high frequency equivalent circuit. Determine the transfer function and the most significant pole assuming $\gamma = 0$ 10 L3 4 3
- b) Determine the maximum noise voltage that a single MOSFET can generate. Also determine the total input-referred noise voltage in a source follower with load implemented using a NMOSFET in saturation. 10 L2 4 3
8. a) Explain the concept of input referred noise voltage. Also calculate input-referred noise voltage for the common source stage with resistive load. 10 L2 4 3
- b) Determine the total output noise voltage in a common source stage with resistive load. 10 L3 4 3

Unit – V

9. a) Determine the gain and frequency of oscillation in each stage of a 3-stage ring oscillator. Also obtain the expression for the output. 10 L2 5 3
- b) Derive the mathematical model of a VCO. Also determine the output and its spectrum in a VCO with control voltage $V_{cont} = V_m \cos \omega_m t$. 10 L2 5 3
10. a) Draw the circuits of one-stage OPAMP and two-stage OPAMP. Briefly explain the same including the expression for the gain of each circuit. 10 L2 5 3
- b) Explain the application of PLL for frequency multiplication and frequency synthesis. 10 L2 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

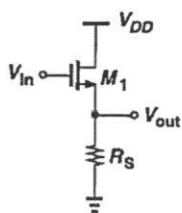


Figure Q.4b

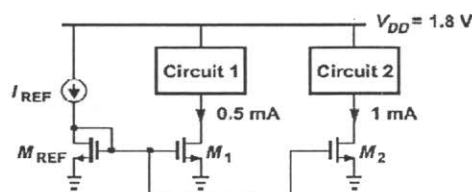


Figure Q.5b

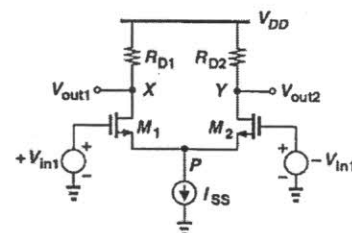


Figure Q.6b.